VIGNESH PRABHU SHANMUGAM

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Portfolio: [Link] | LinkedIn: [Link]

PERSONAL DETAILS

Date of Birth:	26.04.2001
Place of Birth:	India
Nationality:	Indian
Current Location:	76187 Karlsruhe, Germany

EDUCATION

10/2025 – Expected 2027

M.Sc. Mechanical Engineering
Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany
Focus Areas: 1. Computational and Applied Mechanics, 2. Robotics & AI

08/2018 – 07/2022

B.E. Mechanical Engineering
Visvesvaraya Technological University (VTU), India
Aggregate CGPA: 8.86 / 10.00

RESEARCH EXPERIENCE

04/2026 – Current

Research Assistant (Hiwi)

Institute for Microstructure Technology, KIT

- Working on microfluidic mixing enhancement using Koch curve geometries for NMR compatible systems.
- Performing CFD using COMSOL to analyse flow behaviour, pressure drops and mixing efficiency under laminar & time dependant settings.
- Supporting experimental validation and system characterization.

PROFESSIONAL EXPERIENCE

05/2024 – 10/2025

CAE Mechanical & Manufacturing Engineer

TATA Advanced Systems Limited, India

- Performed structural Finite Element Analysis and thermal Computational Fluid Dynamic analysis on aerospace composite materials using Altair Hyperworks, Solidworks, and Abaqus to optimize designs based on performance requirements and manufacturability constraints
- Developed Python scripts that integrated with CFD solvers to simulate composite curing processes, replacing physical manufacturing trials and reducing development costs through digital optimization
- Built VBA [Visual Basics for Application] automation tools for manufacturing information systems, automating data workflows and improving efficiency across engineering and production teams
- Drove digital initiatives by implementing AI-assisted optimization workflows that accelerated design iterations and reduced prototype testing cycles

08/2022 – 05/2024

Assistant Manager (Planning and Technical Procurement)

JSW Steel, LCP 18MT Expansion Project, India

- Led planning and scheduling for major industrial expansion in SMS, implementing process modifications that optimized resource utilization by 15%.
- Managed project execution from conception to commissioning, ensuring adherence to budget and timeline targets
- Handled vendor and contract management, initiating purchase orders in SAP according to technical requirements and statutory policies
- Evaluated supplier proposals and quotes based on technical specifications, coordinating seamless collaboration between teams and contractors

RESEARCH PROJECTS & INTERNSHIP

2025

Physics-Informed AI Digital Twin for Composite Curing | Personal Research Project
Developed an AI-driven digital twin using reinforcement learning (PPO agent) to autonomously optimize curing cycles for thick composite laminates. Integrated physical constraints including tool thermal inertia, epoxy kinetics, and multi-zone heating. 20+ hours of t aerospace specifications, achieving 95% cure uniformity versus 43% with standard profiles through precise temperature control. Reduced simulation time from weeks to hours. **Tools:** Python, TensorFlow, CFD simulation coupling

2022

Research Study – ADI Material Characterization | Bachelor's Thesis | VTU
Examined the behavior of Austempered Ductile Iron (ADI) material, analyzing grain structure formation and response to mechanical deformation. Applied experimental and analytical methods to understand material properties for potential aerospace applications.

2021

Humidity Control in Electrical Environments | Bachelor's Minor Project| VTU
Developed feasible approaches to control relative humidity with minimal maintenance requirements, demonstrating potential to increase component lifespan by 4% (equivalent to 8,000 additional miles in electric vehicles). Project combined materials analysis with practical system design.

2021

SAE India – Hybrid Future Mobility (Efficycle 2020) | Team Octet Captain | Overall Winner, Best CAE Runner-Up
Led interdisciplinary team in national competition for lightweight vehicle design. Managed complete development cycle including conceptual design, FEA simulation, and fabrication. Achieved recognition as overall winner in conventional category nationwide and runner-up for best CAE analysis.

2020

Startup Internship – E-Bike Product Development | Design & Production Intern Engaged in market development for innovative e-bike design. Managed end-to-end product development including concept generation, CAD modeling, prototyping, and production planning from initial idea through to market readiness.

TECHNISCHE FÄHIGKEITEN

- **CAE & Simulation:** Altair Hyperworks (advanced), Solidworks (CFD/MBD), Abaqus, COMSOL
- **3D Modelling:** Solidworks, Autocad, NX
- **Programming & Software:** Python, MATLAB, SAP Material Management, Minitab
- **Core Competencies:** Project management, vendor management, process optimization, technical documentation

SPRACHEN

Englisch:	C1 (IELTS 7.0, geprüft 31.01.2025)
Deutsch:	A2 (Grundkenntnisse, derzeit in Ausbildung)
Others:	Kannada, Hindi, Tamil (Muttersprache)